

Capital Rationing

Capital Rationing

- The practice of restricting capital expenditures to a certain amount
 - reluctance to incur increasing levels of debt
 - limits on external financing
 - management may not want to add to equity in fear of diluting control
 - capital rationing limits a firm in terms of achieve its maximum value

Capital Constraints

- When projects are independent, and the firm has few constraints on capital:
- Ensure that projects at least meet a minimum criteria for acceptance.
- $NPV \geq 0$; $IRR \geq \text{hurdle rate}$; $PI \geq 1$
- If the firm has capital rationing, then its funds are limited and not all independent projects may be accepted.
- Select those projects that best use the firm's available funds.
- If the capital constraint is removed, then all projects can be accepted together.

Market Factors

- Capital rationing may arise due to external factors like imperfections of capital market or deficiencies in market information.
- Either the capital market itself or the Government will not supply unlimited amounts of investment capital to a company, even though the company has identified investment opportunities which would be able to produce the required return.
- These imperfections imply that the firm may not get necessary amount of capital funds to carry out all the profitable projects.

Firm Factors

- Reluctance to take resort to financing by external equities in order to avoid assumption of further risk.
- Reluctance to broaden the equity share base for fear of losing control.
- Reluctance to accept viable projects because of its inability to manage the firm in the scale of operation resulting from inclusion of all the viable projects.

Deciders

- Top management philosophy towards capital spending.
- Outlook for future investment opportunities that may be unavailable if extensive current commitments are undertaken.
- Funds provided by current operations.
- Feasibility of acquiring additional capital through borrowing or share issues.

Method

- Determine profitable investment opportunities
- Decide to obtain that combination of the profitable projects which yields highest NPV within the available funds by ranking them according to their relative profitability.
- Theoretically, projects should be undertaken to the point where the return is just equal to the cost of financing these projects.
- If safety and control are considered than additional profits, there may be a marked unwillingness to engage in external financing, and hence a limit will be placed on the amounts available for investment.
- Even though the firm may wish to raise external finance for its investment, there are many reasons (such as risk) as to why it may be unable to do this.

Example

- Assuming the cost of capital as 10% and the firm is constrained to only invest 500 now

Year	Project A	Project B	Project C	Project D	Project E
0	-500	-200	-200	-200	-100
1	600	242	-100	250	126
2			390		

Example Evaluation

Year	Project A	Project B	Project C	Project D	Project E
NPV	45	20	31	27	15
IRR	9%	10%	6%	14%	15%
PI	1.09	1.10	1.11	1.14	1.15
Rank					
NPV	1	4	2	3	5
IRR	4	3	5	2	1
PI	5	4	3	2	1

Project Selection

Year	Project A	Project B	Project C	Project D	Project E
0	-500	-200	-291	-200	-100
Rank					
NPV	1	4	2	3	5
IRR	4	3	5	2	1
PI	5	4	3	2	1

Firm is to only invest 500 now

Project Selection

Year	Project A	Project B	Project C	Project D	Project E
0	-500	-200	-291	-200	-100
Rank					
NPV	1	4	2	3	5
IRR	4	3	5	2	1
PI	5	4	3	2	1

Firm is to only invest 600 now

Project Selection Implications

- Select Projects A and E (NPV Criteria)
- Select Projects B, D and E (IRR Criteria)
- Select Projects C, D and E (PI Criteria)

Criteria	NPV	IRR	Total Investment	Cost	Adjusted NPV	Adjusted IRR
NPV (A,E)	60	10.00%	600	0	60	10.00%
IRR (B,D,E)	62	10.30%	500	10	52	8.64%
PI (C,D,E)	73	10.00%	591	0.9	72.1	9.85%

Cost = Unutilized investment x cost of capital

Adjusted NPV = NPV – Cost

Adjusted IRR = $\sum_{i=1}^n IRR_i \times w_i$; for 1 to n projects, w_i is the weight of investment on total funds

Soft Rationing

- Many firms' capital constraints are "soft."
- They reflect no imperfections in capital markets.
- Capital rationing are provisional limits adopted by management as an aid to financial control.
- Some ambitious divisional managers habitually overstate their investment opportunities.
 - Rather than trying to distinguish which projects are investible, top management may find it simpler to impose an upper limit on divisional expenditures and thereby force the divisions to set their own priorities.
 - In such instances budget limits are a rough but effective way of dealing with biased cashflow forecasts.
- Management may believe that very rapid corporate growth could impose strains on management and the organization.
 - Since it is difficult to quantify such constraints explicitly, the budget limit may be used as a proxy.
- Soft rationing should never cost the firm anything.

Hard Rationing

- Heavy cost of fund procurement requires hard rationing.
- Capital constraints become tight when the firm raises more money and loosens the investment opportunity.
 - Projects with significant positive NPVs are declined investment funding
- Hard rationing implies market imperfections.
 - A barrier between the firm and capital markets
 - Firm's shareholders lack free access to a functioning capital market
- Market imperfections restrict shareholders' portfolio choice.
- NPV rule assumes that the firm can maximize shareholder wealth by accepting every project that is worth more than it costs.
- However hard rationing deny the firm the market access and hence the cost of equity can not be a representation of the cost of capital.

Linear Programming Model

- Linear programming models solve capital budgeting problems when resources are limited.
- Even if capital is not rationed, other resources may be.
 - Availability of management time, skilled labor, or even other capital equipment often constraint a firm's growth.
- LP models could become complex.
- LP models assume that all future investment opportunities are known.

Example

- Firm is faced with the set of eight investments projects.
- Firm limits the total capital expenditure to 40 million and 20 million in each of the next two years, respectively.
- Select the combination of investments that provides the largest possible return (NPV) without violating either of the two constraints on total capital expenditures.
- The general maximization formula for this problem:
$$\beta_1 Proj_1 + \beta_2 Proj_2 + \beta_3 Proj_3 + \beta_4 Proj_4 + \beta_5 Proj_5 + \beta_6 Proj_6 + \beta_7 Proj_7 + \beta_8 Proj_8 = Max$$

Information

Linear Programming Problem Formulation

Project	Investment Year 1	Investment Year 2	NPV
1	12	3	14
2	54	7	17
3	6	6	17
4	6	2	15
5	30	35	40
6	6	6	12
7	48	4	14
8	36	3	10
Limited	40	20	Max

R2	P1	P2	P3	P4	P5	P6	P7	P8	
Max	14	17	17	15	40	12	14	10	
R1	12	54	6	6	30	6	48	36	≤ 40
R2	3	7	6	2	35	6	4	3	≤ 20

Example

	Results
Maximized NPV	63.0000
Constraint1	40.0000
Constraint2	20.0000
Project 1	1.0000
Project 2	0.0000
Project 3	1.0000
Project 4	1.0000
Project 5	0.0667
Project 6	1.0000
Project 7	0.1667
Project 8	0.0000

Exercise

Year	0	1		NPV
Project				
A	-10000	-10000		6700
B	0	-20000		9000
C	-10000	0		0
D	-15000	0		-1500
Constraint	20000	20000		